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Modeling of Skid-Steer Mobile Manipulators Using Spatial Vector Algebra and Experimental Validation with a Compact Loader

Sergio Aguilera[1], Miguel Torres-Torriti[1] and Fernando Auat[2]

Abstract— **The present work models the dynamics of general skid-steer mobile manipulators using the formalism and tools of the spatial vectors algebra introduced by Featherstone. The model built is validated using inertial measurements obtained during field tests with a compact skid-steer loader. The paper demonstrates the benefits of using the spatial vector algebra formulation, showing that this modeling approach allows to integrate traction forces and study the arm-vehicle, as well as vehicle-ground interactions in a single model. This feature is not possible with many other of the existing modeling approaches and simulation tools, thus opens the way to research on mechanically more complex robot designs and their controllers. It is to be noted that most of the existing models and simulations of mobile manipulators consider two-wheeled differentially driven bases and avoid accurate models of skid-steering bases because of the complexity of simulating wheels that skid while rolling. However, skid-steer traction is common in most of the industrial construction and mining machinery because of their simpler mechanics, high reliability, and better mobility in rough terrains. Hence, the development of physically accurate models of skid-steer manipulators is fundamental. We chose to validate the model using a Cat® 262C compact-skid steer loader instead of a small mobile manipulator common in robotics research laboratory to highlight the usefulness of the presented model and the spatial vector algebra approach.**

I. INTRODUCTION

Skid-steer mobile manipulators (SSMMs) are the integration of a robotic arm along with a skid-steer mobile base. These two elements combine the dexterity of the manipulator to interact with the environment and the mobility of the base to achieve an unlimited workspace. Furthermore, the skid-steer main advantages over other drive-mechanisms are their simpler mechanics, high reliability, and better mobility in rough terrains. In spite of these advantages, they entail additional complexity in the kinematic and dynamic model.

However these advantages, entail additional complexities in the kinematic and dynamic model that arise from such issues the possibly redundant degrees-of-freemdom (DOFs), the skidding effects and the vehicle-manipulator interaction in which ground contact forces are propagated from the base to the arm, while the moving arm causes shifts in the center of mass (COM) of the robot and the propagation of forces back onto the mobile base. Because of this complexity few results exist in the published literature concerning SSMMs. To the best of our knowledge, only the work by Liu et al. [1] treats the modeling of SSMMs with some detail. Lie et al. present a kinematic and dynamic model of SSMMs that takes into account traction and skidding forces, as well shifts in the COM of the robot due to changes in arm position. However, the proposed model does not consider explicitly the propagation of ground interaction forces to the arm, nor the propagation of arm-accelerations back to the base. The model in [1] assumes the mobile base moves in a 2D plane, therefore is not a fully 6-DOF floating base with ground contact interactions. Due to the lack of works treating the modeling of SSMMs, we discuss next the existing research on mobile manipulators and skid-steer vehicles.

Different models for mobile manipulators have been developed over the last years, such as the ones proposed by [2], [3] that consider Ackermann steering geometry, or [4], [5], [6], which consider differential-drive schemes with caster wheels. In general, these models separate the vehicle dynamics from that of the manipulator and do not consider the effects of the moving arm over the base trajectory. This is usually because the arm mass and velocity are assumed to be negligible, or because they focus on problems of redundancy resolution and trajectory planning. Thus for the physically accurate modeling of mobile manipulators it is desriable to establish unified dynamical models that simultaneously consider the coupled interaction between the base and the manipulator.

Regarding skid-steer base models, the main difficulty is the modeling of the lateral skidding of the wheels, produced by the different velocities between the right and left wheels. Some model the skid-steer as a differential drive base in which the lateral slippage may be neglected [7], while others take into account only longitudinal slipping [8] or more detailed lateral skidding models [9], [10].

In general, a weakness of many of the published models is that they are only simulated for controller development purposes and very few of them experimentally validate their models. Some of these exceptions are found in [5], [7], [11], which conduct experimental verifications using small to medium size robots like like Pioneer P3-AT® used by several research groups. It is however desirable from an application perspective to validate the models also with large size and heavier industrial machinery.

The goal of the current research is to present a dynamical model for SSMMs that is general enough and jointly takes into account the coupled interaction between the mobile base and the manipulator, as well as the vehicle-ground interaction in a single and unified fashion. To this end, we rely on the modeling approach introduced by Featherstone using spatial vector algebra [12] for rigid body dynamcis. The spatial

[1]Department of Electrical Engineering, Pontificia Universidad Católica de Chile, Av. Vicuña Mackenna 4860, Macul, Santiago, Chile. sfaguile@uc.cl, mtorrest@ing.puc.cl
[2]Department of Electronic Engineering, Universidad Técnica Federico Santa María, Av. España 1680, Valparaiso, Chile. fernando.auat@usm.cl

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vector formalism allows to model complex kinematic trees with compact equations and a high degree of generality. The model is built using the Spatial Toolbox for Matlab [13] and provides a nontrivial and enriching example that illustrates the capabilities of the modeling approach and the toolbox applied to a real world robot. This paper also validates the model using data experimentally acquired from a compact skid-steer loader Cat® 262C Series 2, which is a good representative of similar machines employed in construction and mining. The model developed and measurements have been made publicly available at [14] for other researchers and students.

The paper is organized as follows. In section II we present the dynamic modeling of a general SSMM using spatial vector algebra. Section III describes the simulation details, the testing methodology, as well as the analysis of the experimental and simulated results. Finally, section IV presents the conclusions of our work and discusses some aspects concerning ongoing research.

II. Modeling of an SSMM Using Spatial Vector Algebra

This section discusses the construction of a *complete* and *general* dynamic model of an SSMM as the one shown in fig. 1. By *complete* we mean the model considers (i) a floating mobile base which is not restricted to move on a 2D plane as in most models described in the literature, and (ii) includes the ground interaction forces at the contact points of the mobile base. By *general* we mean that the model can respresent any SSMM by adequate choice of the geometric and inertial parameters. To this end, we employ the spatial vector algebra approach for rigid body dynamics proposed by Featherstone [12]. The main reasons for this choice over conventional modelings are the fewer equations and higher level of abstraction, while at the same time provides valuable insight into the dynamics and physical properties of multibody robotic systems. We next explain how to build a general SSMM model using the spatial coordinate transforms that are one of the basic ingredients of the spatial vector algebra approach and how the traction forces are introduced.

A. General SSMM Model

To build the model it is convenient to first define the bodies and joints of the robot. To this end, the Cartesian coordinate frame $\mathcal{F}_0$ is first placed at a chosen fixed location in space to serve as virtual fixed base (i.e. global inertial frame) [cf. Fig. 1]. Next the mobile base is labeled as body 1 with coordinate frame $\mathcal{F}_1$. A convenient location for $\mathcal{F}_1$ is the robots center of mass. The mobile base (body 1) is treated as a body connected to the fixed base (body 0) by a six-degree-of-freedom (6-DOF) joint, i.e. the mobile base is a *floating base* allowed to move freely without any kinematic constraints save for non-permanent ground contact constraints. Attached to the mobile base are the wheels connected to the base by rotary joints. The wheels are bodies labeled 2, 3, 4 and 5 with corresponding Cartesian frames $\mathcal{F}_i$, $i = 2, 3, 4, 5$ as shown in fig. 1. Similarly, the robot arm is a series of bodies with coordinate frames $\mathcal{F}_i$, $i = 6, 7, 8, \dots, N$, where N is the last body of the arm and represents also the total number of bodies of the robot. The connectivity graph for the N bodies of the robot is shown in fig. 2. The nodes of the graph represent each body, while the lines connecting the nodes represent the joints of the robot, such that joint i is the joint that connects body i to its parent. The SSMM is a kinematic tree, hence its connectivity graph is a topological tree. Adding more arms or wheels would add additional branches to the tree in fig. 2.

Having numbered the bodies and joints, the connectivity of the robot can be completely described by an array $\lambda \in \mathbb{Z}^{*N}$, such that $\lambda(i)$ (the i-th entry of the array), contains the body number of the parent of body i. From the connectivity graph of fig. 2 it should be clear that $\lambda = [0, 1, 1, 1, 1, 1, 6, 7, \dots, N-1]$. In addition to the connectivity of the robot bodies, to complete the geometric description of the robot it is necessary to define the geometric transformations relating the location of each joint relative to the reference frame of the body to which each joint is attached. Formally, this requires first to introduce pair of coordinate frames for each joint i that links body i to its parent $\lambda(i)$. One frame is labeled $\mathcal{F}_i$ and fixed to the body i, while the other is labeled $\mathcal{F}_{\lambda(i),i}$ is fixed to the parent body $\lambda(i)$. To minimize the number of paremeters required to describe the relative motion between $\mathcal{F}_i$ and $\mathcal{F}_{\lambda(i),i}$ it is convenient to locate the frames such that both frames coincide when the joint variables are zero, and have axes aligned following a set of rules like the widely employed Denavit and Hartenberg convention to constrain the possible frame locations [12]. Other conventions than the D-H procedure to define coordinates frame are posible. In fact, the spatial vector algebra approach does not require frames to be defined according to the D-H procedure. However, using

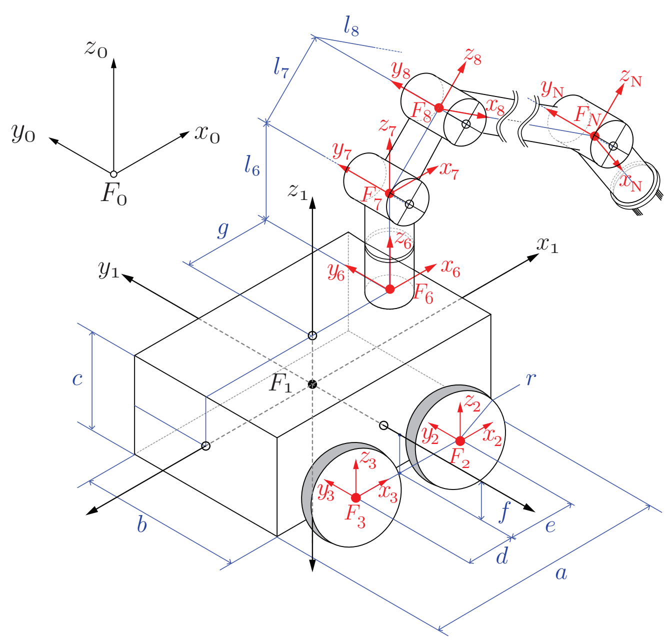

Fig. 1. Skid-steer mobile manipulator.

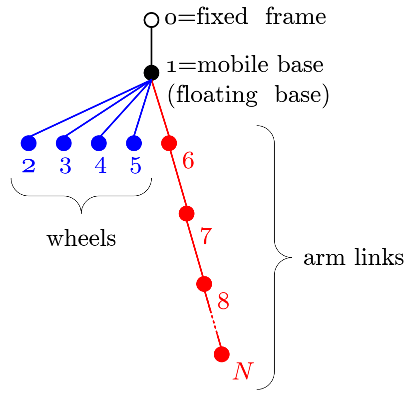

Fig. 2. Kinematic tree of the skid-steer mobile manipulator of fig. 1.

D-H procedure as part of the spatial vector algebra approach reduces the number of joint paremeters to a minimum of four with one of the parameters acting as joint variable. Noting that each body i contains a frame $\mathcal{F}_i$ and a variable number of frames $\mathcal{F}_{\lambda(j),j}$, for all j satisfying $\lambda(j) = i$ (with frames $\mathcal{F}_{\lambda(j),j}$ located at the joints of the children bodies j whose parent body is body $\lambda(j) = i$), it is convenient to select frame $\mathcal{F}_i$ as the coordinate system for body i in terms of which will be defined the spatial inertia of body i. Finally, a complete description of the robot geometry is obtained defining two transformations: $\mathbf{X}_T(i)$ and $\mathbf{X}_J(i)$. The transformation $\mathbf{X}_T(i)$ is the Plücker coordinate transform from body $\lambda(i)$ coordinates in frame $\mathcal{F}_{\lambda(i)}$ to the coordinates in frame $\mathcal{F}_{\lambda(i),i}$ located at joint i, but fixed to body $\lambda(i)$. The coordinate transform $\mathbf{X}_J(i)$ is the joint coordinate transform mapping coordinates from frame $\mathcal{F}_{\lambda(i),i}$ to coordinates in the frame $\mathcal{F}_i$ fixed to the children body i of parente body $\lambda(i)$. Therefore, with these transforms it is possible to construct the so-called *link-to-link* transform

$$ {}^{i}\mathbf{X}_{\lambda(i)} \quad = \quad \mathbf{X}_J(i)\mathbf{X}_T(i) \tag{1} $$

from coordinates in frame $\mathcal{F}_\lambda(i)$ of body $\lambda(i)$ to coordinates in the coordinates of frame $\mathcal{F}_i$ in body i. The complete set of transformations $\mathbf{X}_T(i)$, $i = 1, 2, \ldots, N$ describes the location of each joint-frame $\mathcal{F}_{i,j}$ realtive to its corresponding body-frame $\mathcal{F}_i$ within body i. The body constant geometry data contained in $\mathbf{X}_T(i)$, $i = 1, 2, \ldots, N$, together with the variable joint coordinate transforms $\mathbf{X}_J(i)$, $i = 1, 2, \ldots, N$ and the connectivity data in the parent array λ, permit to completely describe the geometry of the robot and the position of its bodies with respect to the global reference frame $\mathcal{F}_0$. The recursive formula based on the link-to-link transformation (1):

$$ {}^{i}\mathbf{X}_0 \quad = \quad {}^{i}\mathbf{X}_{\lambda(i)}\, {}^{\lambda(i)}\mathbf{X}_0, \quad \text{with } \lambda(i) \neq 0. \tag{2} $$

allows to compute the coordiante transform from the global reference frame $\mathcal{F}_0$ to the body coordinate frame $\mathcal{F}_i$, thus allowing to solve the forward kinematics. Equations (1) and (2) are also at the base of recursive inverse kinematics and forward/inverse dynamics computations with algorithms whose detailed discussion can be found in [12]. It should be stressed that the transforms (1) and (2) are the basic building block for the model of any mechanism. The specific transformations required to build the SSMM model are summarized in Table I together with the joint variables.

From Table I it is possible to observe that the mobile base is a free-floating body with 6-DOF (three for orientation and three for translation). However, to avoid the singularities of Euler Angles (roll, pitch, yaw), the rotations are expressed in term of Euler Parameters (unit quaternion) involving an axis of rotation (u_x, u_y, u_z) and an angular amount θ that give rise to four unit quaternion parameters (p_0, p_1, p_2, p_3). Therefore, the description of the state of the first body involves a thirteen-dimensional vector:

$$ \mathbf{x} = [\underbrace{p_0, p_1, p_2, p_3}_{\substack{\text{Orientation}\\ \text{quaternion}}}, \underbrace{q_x, q_y, q_z}_{\substack{\text{Position}\\ \text{relative}\\ \text{to } \mathcal{F}_0}}, \underbrace{\omega_x, \omega_y, \omega_z}_{\substack{\text{Angular}\\ \text{velocity in } \mathcal{F}_1\\ \text{coordinates}}}, \underbrace{v_x, v_y, v_z}_{\substack{\text{Linear}\\ \text{velocity in } \mathcal{F}_1\\ \text{coordinates}}}] $$

For the remaing 1-DOF joints for the wheels and the manipulator only the angular position q_i and angular velocity $\dot{q}_i$, $i = 2, 3, \ldots, N$ are required to complete the description of the state of the bodies. Thus for a four-wheeled SSMM with an M degrees of freedom manipulator, the full state vector would be given by $\mathbf{q}_{SSMM} = [\mathbf{X}|q_2 \; q_3 \cdots q_{M+5}|\dot{q}_2 \; \dot{q}_3 \cdots \dot{q}_{M+5}]$, requiring $13 + 4 \times 2 + M \times 2 = 13 + 2 \times (M + 4)$ joint variables.

B. Traction Force

The main characteristics that are modeled for traction forces and vehicle-ground interactions are damping (D), stiffness (K) and friction (μ). The applied force appears over the surface in contact with the ground. To this end, each body has a set of pre-defined contact points (CPs) declared in their own frame. Since not every CP is in contact with the ground at all times, the contact state of each CP must be verified before a force is applied. Defining the distance between the i-th CP and the ground surface as:

$$ \delta_i = n_{x_i,y_i}^T \, p'_i - d_{x_i,y_i}, \tag{3} $$

where $p'_i = p_i - [x_i, \; y_i, \; (d - n_{xi}x_i - n_{yi}y_i)/n_{zi}]^T$ and $p_i = (x_i, y_i, z_i)$ is the position of the i-th CP in the global frame. The vector p'_i is the position of the i-th CP relative to it expected ground position that represents the deformation between the rigid body point and the compliant surface. The vector n_{x_i,y_i} is the surface normal vector at location (x_i, y_i) and d_{x_i,y_i} is the distance of the surface plane to the origin for a ground surface described as a piecewise continuous concatenation of planes. If $\delta_i \leq 0$, the i-th CP is in contact and the magnitude of the normal force is calculated as:

$$ N_i = \sqrt{-\delta_i}\left[-K\delta_i - D\dot{\delta}_i\right], \tag{4} $$

while the magnitude of the force tangent to the surface is given by $f_{ti} = \mu N_i$ as long as $|\mu N_i| < |f_{stick}|$, otherwise $f_{ti} = f_{stick}$, whenever the sticking force is exceeded. The sticking force magnitude is defined as $f_{stick} = -K_t\epsilon_i - D\dot{\epsilon}_i$, where $\epsilon_i = \|p'_i - n_{x_i,y_i}^T \, p'_i \, n_{x_i,y_i}^T\|$ is the magnitude of the projection of p'_i onto the surface plane.

TABLE I

SSMM MODEL PARAMETERS USING SPATIAL VECTOR ALGEBRA.

Joint-i	1	2,3,4,5	6	7, 8, …, N
Type	Floating	Rotary $\mathbf{y}_i, i = 2, 3, 4, 5$	Rotary $\mathbf{z}_6$	Rotary $\mathbf{y}_i, i = 7, 8, \ldots$
Body-i	1	2,3,4,5	6	7, 8, …, N
Description	Mobile Base	Wheels	First Arm Link	Remaining Arm Links
Parent Body $\lambda(i)$	0	1	1	6–$(N-1)$
Dimensions				
x	a	$2r$	ϵ	ϵ
y	b	w	ϵ	ϵ
z	c	$2r$	l_6	l_i
Joint-location Transforms $\mathbf{X}_T(i)$	$xlt([0, 0, 0])$	$i = 2: \ xlt([e, -b/2, -f])$, $i = 3: \ xlt([-d, -b/2, -f])$, $i = 4: \ xlt([e, b/2, -f])$, $i = 5: \ xlt([-d, b/2, -f])$	$xlt([g, 0, c/2])$	$xlt([0, 0, l_{i-1}])$
Joint Transforms $\mathbf{X}_J(i)$	$rotu(\cdot)xlt(\mathbf{E}_u^{-1}[q_x, q_y, q_z])$	$roty(q_2)$, $roty(q_3)$, $roty(q_4)$, $roty(q_5)$	$rotz(q_6)$	$roty(q_i)$
Mass	m_1	$m_2 = m_3 = m_4 = m_5$	m_6	m_i
COM in $\mathcal{F}_i$ coordinates	$[0, 0, 0]$	$[0, 0, 0]$	$[0, 0, l_6/2]$	$[0, 0, l_i/2]$
Inertia about COM	$\frac{m_1}{12}\begin{bmatrix} b^2+c^2 & 0 & 0 \\ 0 & a^2+c^2 & 0 \\ 0 & 0 & a^2+b^2 \end{bmatrix}$	$\frac{m_i}{12}\begin{bmatrix} 3r^2+w^2 & 0 & 0 \\ 0 & 6r^2 & 0 \\ 0 & 0 & 3r^2+w^2 \end{bmatrix}$	$\frac{m_6}{12}\begin{bmatrix} 3\epsilon^2+l_6^2 & 0 & 0 \\ 0 & 3\epsilon^2+l_6^2 & 0 \\ 0 & 0 & 6\epsilon^2 \end{bmatrix}$	$\frac{m_i}{12}\begin{bmatrix} 3\epsilon^2+l_6^2 & 0 & 0 \\ 0 & 3\epsilon^2+l_6^2 & 0 \\ 0 & 0 & 6\epsilon^2 \end{bmatrix}$

Summary of 3D arithmetic functions (cf. [12] for a complete list.)

$$\mathbf{E}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & s \\ 0 & -s & c \end{bmatrix}, \ \mathbf{E}_y(\theta) = \begin{bmatrix} c & 0 & -s \\ 0 & 1 & 0 \\ s & 0 & c \end{bmatrix}, \ \mathbf{E}_z(\theta) = \begin{bmatrix} c & s & 0 \\ -s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{v}\times = \begin{bmatrix} 0 & -v_z & v_y \\ v_z & 0 & -v_x \\ -v_y & v_x & 0 \end{bmatrix}$$

with $c = \cos(\theta), s = \sin(\theta), \mathbf{v} = [v_x, v_y, v_z]$;

$$\mathbf{E}_u(\theta) = \begin{bmatrix} p_0^2 + p_1^2 - 1/2 & p_1p_2 + p_0p_3 & p_1p_3 - p_0p_2 \\ p_1p_2 - p_0p_3 & p_0^2 + p_2^2 - 1/2 & p_2p_3 + p_0p_1 \\ p_1p_3 + p_0p_2 & p_2p_3 - p_0p_1 & p_0^2 + p_3^2 - 1/2 \end{bmatrix} \text{ (rotation about axis } u = [u_x, u_y, u_z]);$$

with Euler parameters (unit quaternion) $p_0 = \cos(\theta/2), p_1 = \sin(\theta/2)u_x, p_2 = \sin(\theta/2)u_y, p_3 = \sin(\theta/2)u_z, p_0^2 + p_1^2 + p_2^2 + p_3^2 = 1$.

Summary of spatial vector arithmetic functions (cf. [12] for a complete list.)

$$xlt(\mathbf{r}) = \begin{bmatrix} \mathbf{1}_{3\times3} & \mathbf{0}_{3\times3} \\ -\mathbf{r}\times & \mathbf{1}_{3\times3} \end{bmatrix}, \ rot*(\theta) = \begin{bmatrix} \mathbf{E}_*(\theta) & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{E}_*(\theta) \end{bmatrix}, \ * = x, y, z, u$$

III. Simulation and Experimental Verification

A. Simulation Details

The Spatial Toolbox (version 2) for Matlab [13] was used to implement the dynamic model of an SSMM. The Spatial Toolbox provides a set of algorithms and functions to simulate the motion dynamics of multibody mechanical systems using the spatial vector algebra approach. The toolbox allows to build the models in terms of easy to use data structures that only require to fill in the parent array λ of the the kinematic structure, the relative position of the joints relative to the parent body (i.e. the parameters of the joint-location transform $\mathbf{X}_T$), the location of the COM of each body, and the mass and inertia matrices of the bodies. The toolbox contains functions to compute the forward and inverse dynamics that can be called from Matlab scripts or from Simulink simulation models. The functions provided also allow to include user-defined contact and motion constraints. The results from the simulations can be visualized using functions that are part of the toolbox and that allow to create 3D graphical representations of the robot or mechanical system.

The SSMM model built for the simulations corresponds the one presented in section II, but with a simplified one-DOF manipulator to provide a realistic representation of a Cat® 262C Series 2 compact skid-steer loader that was robotized for teleoperation and autonomous navigation experiments. Because the arm of the loader moves about an axis equivalent to the $\mathbf{y}_7$-axis of the frame $\mathcal{F}_7$ in the general model of fig. 1, the original joint of body 6 which rotates about $\mathbf{z}_6$-axis is assumed to be fixed. In practice this was implemented by defining a body 6 that represent an arm that can rotate about an axis parallel to axis $\mathbf{y}_1$. Extra body and joint can be easily added to represent the motion of the loader's bucket , however to reduce the number of variables the experiments were carried out using a fixed bucket positon and therefore bodies $i = 7, 8, \ldots, N$ of the general SSMM model were not defined in the model script that can be obtained from [14]. The specific values for the geometric and inertial parameters in Table I needed to model the Cat® 262C are summarized in Table II. It is to be noted that the joint location parameter g of the loader arm corresponding to frame $\mathcal{F}_6$ in fig. 1 shown in Table II is negative. This is because the Cat 262C arm is positioned at the rear-end of the machine, opposite to the front location of the arm in the general SSMM model shown in fig. 1.

TABLE II
CAT® 262 MODELED PARAMETERS IN SI UNITS.

Description		Mobile Base	Wheels	Manipulator
Dimensions	x	$a = 3$	$2r = 0.9$	$\epsilon = 0.15$
	y	$b = 1.6$	$w = 0.25$	$\epsilon = 0.15$
	z	$c = 1.2$	$2r = 0.9$	$l_6 = 3.3$
Mass		$m_1 = 2389$	$m_{2,3,4,5} = 47.7$	$m_6 = 1034$
Joint Location Parameters		–	$d = 0.2$ $e = 1$ $f = 0.5$	$g = -1.2$

Once each body is declared, the ground contact points are defined with respect to each body's frame $\mathcal{F}_i$ in 3D coordinates. These points cannot penetrate the ground plane defined at $z_0 = 0$ in the global reference frame $\mathcal{F}_0$. More specifically, eitght CPs are defined at each corner of the mobile base represented by a rectangular box, one CP is defined at the end of the manipuator arm where the tip of the bucket is located, and each of the four wheels has 32 CPs distributed around its perimeter. The larger number of contact points on the wheels was choosen to have a more realistic simulation of the ground-wheel interaction. At the contact point of the wheels, friction forces resulting from the lateral skidding are applied. It is to be noted that the CPs are not permanent contact points, i.e. some of the wheels may loose contact with the ground and no traction or skidding forces are applied in this case. The current model assumes that the thrust force delived by the engine is distributed among the traction forces arising at the wheel-ground contact point, but applied on the tip of each wheel's axle, i.e. applied at the origin of the wheel-frames $\mathcal{F}_i$, $i = 2, 3, 4, 5$, whenever the wheel is in contact with the ground.

The parameters of internal friction and viscous dissipative forces at joints and engine response were experimentally obtained using measurements taken on the Cat® 262C. To this end, three aspects were considered: (i) response time of the machine to step inputs, (ii) viscous friction forces acting on the hydraulic system when the machine reaches steady-state constant velocity motion, and (iii) machine arm response time of the hydraulic actuator. Concerning the step inputs, the actual model assumes a ramping period of 0.2 s when sending a non-zero velocity set-point and a 0.1 s when sending a zero velocity (stopping) set-point. These ramping periods are included to take into account the actual ramping strategy implemented on the machine for safety reasons and to avoid possible mechanical wear-off and damages from sudden braking. To model the dissipative and viscous friction forces that causes the net acceleration to become zero when the machine moves at constant velocity we considered the following standard approximation, in which the effect of dissipative forces is given by $F = -(C_{vi}v_x + C_{\omega i}\omega_z)$ for each wheel $i = 2, 3, 4, 5$, where v_x is the longitunal velocity of the machine along $\mathbf{x}_1$-axis and ω_z is the turning angular velocity about the $\mathbf{z}_1$-axis.

To estimate the machine arm time response contants we measured the time it takes the arm to fall down when the hydraulic valves are released (i.e. no force is applied to the piston). A PD controller was added to the simulations to control the the arm to a specified position. The applied torque computed by the PD controller is calculated according to

$$\tau(q_{6_e}) \quad = \quad K_p \cdot q_{6_e} + K_d \cdot \frac{d}{dt} q_{6_e}$$

where $q_{6_e} = q_6{}^r - q_6$ is the error between the referecen position $q_6{}^r$ and the measured joint position q_6. The controller constants were set to

$$K_p \;=\; \left(\frac{l_6}{2} m_6 g + l_6 m_{load} g\right), \quad K_d \;=\; 5 K_p T.$$

The chosen value for K_p is equal to the torque that should be applied by the arm at an horizontal equilibrium position while holding a load of size m_{load}. The derivative constant K_d is proportional to K_p and the maximum sampling period $T = 0.003$ s. The term $5T$ is much smaller than the response time of the arm. This ensures a less aggressive controller response that does not compromise the stability of the arm and dampens arm oscillation that affect the vehicle's displacement.

A simplified simulation flowchart is shown on Fig. 3. The simulation starts with the initial conditions for the joint states (position and velocities contained in x, q and $\dot{q}$), the dissipative forces represented by F_d, the external forces including CP forces represented by F_{ext}, the user-supplied setpoints for the mobile base displacement (F_{Linear}, $F_{Angular}$) and the arm reference position q^r. With these data the state of the SSMM is updated by solving the forward dynamics with Featherstone's articulated rigid-body algorithm. Next the wheel forces, ground CP forces and controller signals are computed to be used in the next iteration.

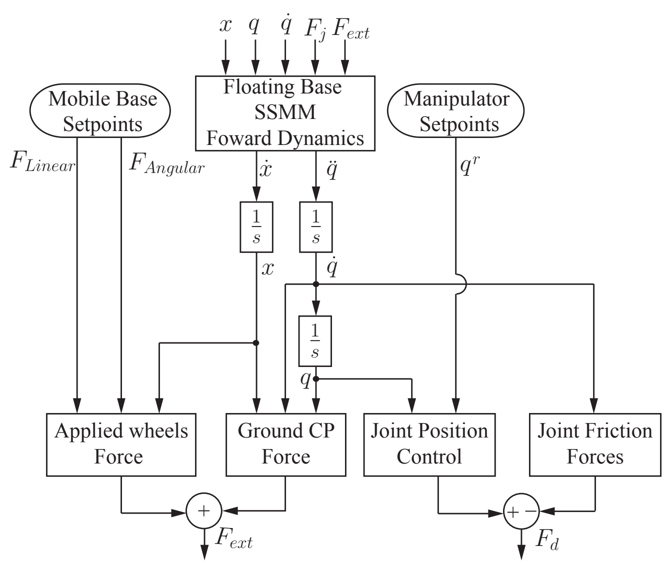

Fig. 3. Simulation flowchart.

B. Field Experiemnts

In order to validate the SSMM model, motion experiments were conducted using the Cat® 262C skid-steer loader shown in fig. 4. The tests were conducted on asphalt covered section of a parking lot with the machine being remotely operated. The tests consisted of two driving maneuvers (i) straight line motion of approximately 4 m, and (ii) in-place 360° turns carried out with and without load. The load applied to the bucket corresponds to 5×80 liter drums of water totaling 400 kg (approx.). This load is equivalent to roughly 11% of the unloaded machine weight and is sufficient to considerably affect the location of the loader's COM. The four set of experiments were repeated ten times each. The acceleration and heading data was acquired using a high precision IMU and gyroscope Crossbow® model RGA300CA. This device outputs linear accelerations in the X, Y and Z axes, angular velocity ω_z, roll and pitch angles ϕ and θ, respectively.

Fig. 4. Compact skid-steer loader at the experiment site with unloaded bucket (left) and loaded bucket (right).

C. Simulation and Experimental Results

The measurements for both straight line motion and in-place rotation show a reasonable agreement with the simulated values. This can be confirmed from fig. 5, which shows the machine position for the twice integrated acceleration measured in the straight line motion experiment without load (blue lines) and the simulated value for the machine longitudinal position (red line). Similarly, the in-place rotation experiment, whose results are shown in fig. 6, display great consistency with the simulated turning rate of the machine. From the straight line experiments and simulation of fig. 5 it is also possible to see that the both the measured and simulated velocities are consistent. In the case of the turning speed shown in fig. 6 the agreement between the measured and simulated acceleration and deceleration is clear. The mismatch in the final linear and angular displacements is due to the fact that setpoints were manually issued, and there is a ±1 second difference in triggering the stopping command.

The experiments carried out with load show that the machine's acceleration was reduced. This can be observed in the straight line motion experiment of fig. 7, which shows that the position curve has a slightly smaller slope when compared to the same curves in fig. 5. Similarly, the in-place turn with the loaded bucket took about 4 seconds more to complete the 360° turn, and the measured turning velocity decrease from roughly 37±2 $^\circ/s$ to 27±2 $^\circ/s$ as seen from the comparison of figs. 6 and 8. In the case of the straight line motion experiment with load it is possible to oberve a larger

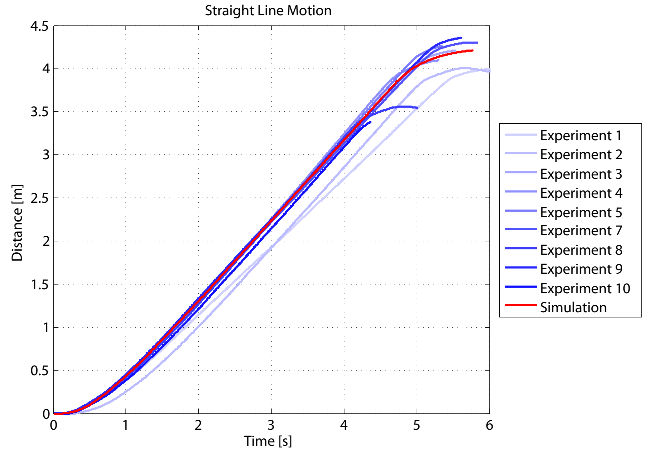

Fig. 5. Straight line motion experimental data v/s simulated model with the SSMM without load.

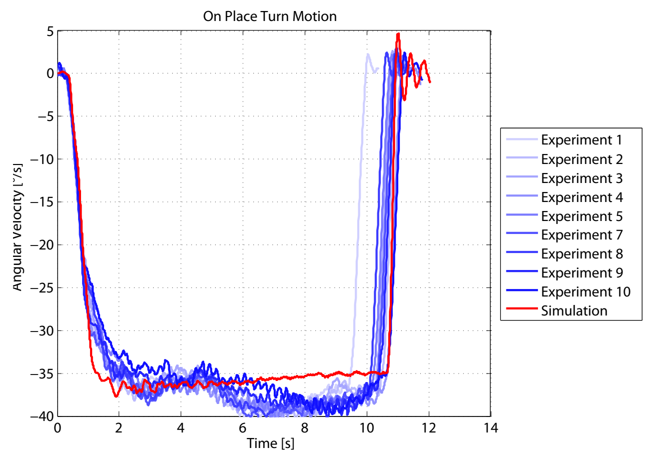

Fig. 6. On place turn motion experimental data v/s simulated model with the SSMM without load.

variation between the finally traveled distances. This larger variation is explained due to the relative decrease of the average measured accelerations with respect to the constant level sensor noise that contaminated the measurements.

It is to be noted that the ripple in the simulated turning speed of the loaded and unloaded machine shown figs. 6 and 8 is produced by the skidding, which is not constant because the CPs change while the machine moves, since there is a discrete number of CPs on each of the 32-sides polygonal wheels, and because of minor arm oscillations due to controller tuning that also induce small disturbances in the CP forces. This effect allows us to appreciate the natural interaction between bodies of the SSMM built using the unified approach of the spatial vector algebra.

IV. Conclusions

A general model for SSMMs considering skidding and vehicle-manipulator interactions was presented. The vehicle-manipulator interactions arise due to COM displacements produced by different manipulator loads and because of

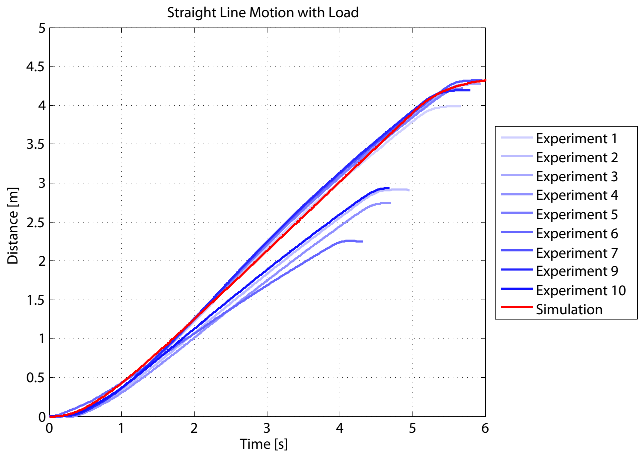

Fig. 7. Straight line motion experimental data v/s simulated model with the SSMM with 400 kg load.

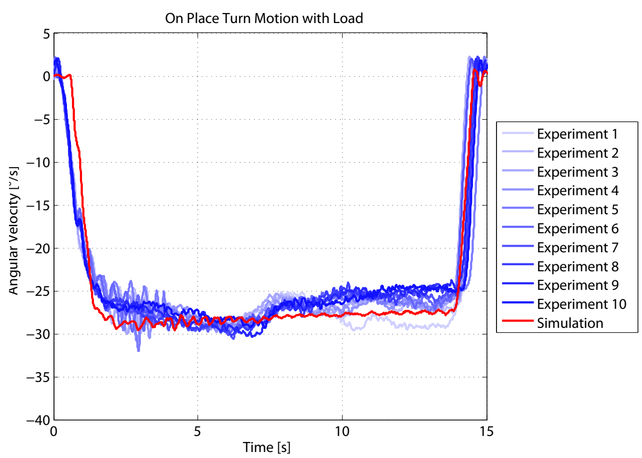

Fig. 8. On place turn motion experimental data v/s simulated model with the SSMM with 400 kg load.

disturbances induced by the arm controller oscillations. The model was experimentally validated by comparing simulation results with measurements acquired from the inertial sensors installed on board of a Cat® 262C compact skid-steer loader. The accuracy between the model and the experimental data reassures the usefulness of the spatial vector formulation and the model built enriches the set of examples included with the Spatial Toolbox. Both the model and experimental data collected during the field tests have been made publicly available at [14] for the robotics community interested in studying the dynamics, motion control and mechanical design of SSMMs, among other related topics. Even though this approach has been validated for a specific machine, it should be easily adaptable to other machines given the generality of the model and provided that the paramters in tables I and II are adequately set.

On going research concerns the refinment of the models that implement the transformation of the machine command signals into a specific engine thrust force and the development of controllers to compensate disturbances propagated across the manipulator due to terrain unevenness. Continuing research is also concerned with enhancing wheel slippage models, and the implementation of autonomous navigation and task execution strategies relying on the built model for SSMMs.

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REFERENCES

[1] Yugang Liu and Guangjun Liu. Modeling of tracked mobile manipulators with consideration of track-terrain and vehicle-manipulator interactions. *Robot. Auton. Syst.*, 57(11):1065–1074, November 2009.

[2] A. De Luca, G. Oriolo, and P. Robuffo Giordano. Kinematic control of nonholonomic mobile manipulators in the presence of steering wheels. In *IEEE Int. Conf. on Robotics and Automation*, pages 1792–1798, Anchorage, AS, USA, May 2010.

[3] J. Tan, N. Xi, and Y. Wang. Integrated task planning and control for mobile manipulators. *Int. J. Rob. Res*, pages 337–354, 2003.

[4] Qing Yu and I-Ming Chen. A general approach to the dynamics of nonholonomic mobile manipulator systems. *Journal of Dynamic Systems, Measurement, and Control*, 124:512 – 521, 2002.

[5] G.D. White, R.M. Bhatt, Chin Pei Tang, and V.N. Krovi. Experimental evaluation of dynamic redundancy resolution in a nonholonomic wheeled mobile manipulator. *Mechatronics, IEEE/ASME Transactions on*, 14(3):349–357, 2009.

[6] Zhijun Li, Shuzhi Sam Ge, and Aiguo Ming. Adaptive robust motion/force control of holonomic-constrained nonholonomic mobile manipulators. *IEEE Transactions on Systems, Man, and Cybernetics, Part B*, 37(3):607–616, 2007.

[7] A. Mandow, J.L. Martinez, J. Morales, J.-L. Blanco, A. Garcia-Cerezo, and J. Gonzalez. Experimental kinematics for wheeled skid-steer mobile robots. In *Intelligent Robots and Systems, 2007. IROS 2007. IEEE/RSJ International Conference on*, pages 1222–1227, 2007.

[8] Jingang Yi, Hongpeng Wang, Junjie Zhang, Dezhen Song, S. Jayasuriya, and Jingtai Liu. Kinematic modeling and analysis of skid-steered mobile robots with applications to low-cost inertial-measurement-unit-based motion estimation. *Robotics, IEEE Transactions on*, 25(5):1087–1097, 2009.

[9] Kozlowski, Krzysztof, Pazderski, and Dariusz. Modeling and control of a 4-wheel skid-steering mobile robot. *International Journal of Applied Mathematics and Computer Science*, 14(4):477–496, 2004.

[10] E. Mohammadpour, M. Naraghi, and M. Gudarzi. Posture stabilization of skid steer wheeled mobile robots. In *Robotics Automation and Mechatronics (RAM), 2010 IEEE Conference on*, pages 163–169, 2010.

[11] Wei Yu, O.Y. Chuy, Jr. Collins, E.G., and P. Hollis. Analysis and experimental verification for dynamic modeling of a skid-steered wheeled vehicle. *Robotics, IEEE Transactions on*, 26(2):340–353, 2010.

[12] Roy Featherstone. *Rigid Body Dynamics Algorithms*. Springer-Verlag New York, Inc., Secaucus, NJ, USA, 2008.

[13] Roy Featherstone. Spatial vector algebra toolboox for matlab. http://royfeatherstone.org/spatial/index.html, 1 February 2014.

[14] S. Aguilera and M. Torres. Experimental raw data and simulation videos. http://ral.ing.puc.cl/ssmm.htm, 5 February 2014.